

**User: Alice** Mar 18

Who organized my longest outdoor activity in March?

Question → **LLM Agent** → Answer

**Query** → **MemCatalog** → **Retrieved memory**

**How to use?**

**How to store?**

## 1. Raw Sessions

| | |
|---|---|
| Mar 2 | User: "Ava arranged our 4-hour outdoor hike at Pine Trail." |
| Mar 9 | User: "Ben booked my 2-hour indoor pottery class." |
| Mar 16 | User: "Cara arranged our 3-hour outdoor ride at Lake Park." |
| Mar 18 | User: Who organized my longest outdoor activity in March?<br>Agent: Ava organized the 4-hour hike. |

**Extract missing facts**

Verify against User spans

## 2. First-order Memory

| activity | property | value | source |
|---|---|---|---|
| hike-1 | setting | outdoor | Mar 2 |
| pottery-1 | setting | indoor | Mar 9 |
| ride-1 | setting | outdoor | Mar 16 |
| hike-1 | organizer | Ava | Mar 2 |
| pottery-1 | organizer | Ben | Mar 9 |
| ride-1 | organizer | Cara | Mar 16 |
| hike-1 | duration | 4 h | Mar 2 |
| ride-1 | duration | 3 h | Mar 16 |

Use / Update

## 3. Second-order Memory

Reported organizer use (Mar 17)

| organizer fact | retrieved | used |
|---|---|---|
| hike: Ava | yes | yes |
| pottery: Ben | yes | no |
| ride: Cara | yes | yes |

Use / Update

| action | cost | outcome |
|---|---|---|
| acquire duration | logged | verified |

## For Accuracy — Query-specific pipelines

### (a) Single-fact retrieval

*Who organized the hike?*

| target |
|---|
| hike-1 |

**Ground** → **Verify**

| subject | predicate | object |
|---|---|---|
| hike-1 | organizer | Ava |

**Select**

| value |
|---|
| Ava |

### (b) Independent composition

*How many March activities were outdoors?*

| activity | setting |
|---|---|
| hike-1 | outdoor |
| pottery-1 | indoor |
| ride-1 | outdoor |

**Evaluate** *outdoor?*

| activity | yes? |
|---|---|
| hike-1 | 1 |
| pottery-1 | 0 |
| ride-1 | 1 |

**Combine** *1 + 0 + 1*

| count |
|---|
| 2 |

### (c) Dependent composition

*Who organized my longest outdoor activity in March?*

| outdoor | duration |
|---|---|
| hike-1 | 4 h |
| ride-1 | 3 h |

**Compare** *4 h > 3 h*

| longest |
|---|
| hike-1 |

**Bind**

| target | property | value |
|---|---|---|
| hike-1 | organizer | Ava |

## For Cost — Cost-aware execution

| work | state | LLM calls |
|---|---|---|
| reuse known facts | verified | 0 |
| hike duration | missing | estimated |
| ride duration | missing | estimated |

Priorities and cost estimates

**Allocate and execute**

Within the query budget

Re-estimate after each action